

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (ISE) (Credit System) Degree Examinations

January 2023

19HU701 – ENGINEERING MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I**Marks**

1. a) Define management. Explain the functions of management. 10
b) Define forecasting. Explain methods of forecasting. 10
2. a) Discuss the various types of organization. 10
b) A business produces 200 units of product by making following expenditure.
i) material – ₹30,000
ii) labour – ₹ 20,000
iii) factory overhead – ₹4,000
iv) administrative overhead - ₹5,754 and
v) selling & distribution overhead. ₹1500
The products are sold at a price of ₹400 per unit.
The above expenditures are classified into fixed and variable as follows.

Expenditure	Fixed	Variable
Materials	Nil	100%
Labour	50%	50%
Factory overhead	25%	75%
Administrative overhead	100%	Nil
Selling & distribution overhead	60%	40%

From the above information determine (a) total variable cost and fixed cost
(b) contribution (c) P/V ratio (d) Break-even- point in units and sales value.

10

3. a) Explain features of sole proprietorship. 10
b) Discuss various legal form of organization. 10

Unit – II

4. a) The following table lists the jobs of a network along with their time estimates:

Job (i - j)	Duration (days)		
	t_o	t_m	t_p
1-2	3	6	15
1-6	2	5	14
2-3	6	12	30
2-4	2	5	8
3-5	5	11	17
4-5	3	6	15
5-8	1	4	7
6-7	3	9	27
7-8	4	19	28

- i) Draw the project network
- ii) Calculate the length and variance of the critical path
- iii) What is the probability that the jobs on the critical path will be completed in 41 days
- iv) What is the probability that the jobs on the next most critical path will be completed by the due date 41 days.

10

P.T.O.

- b) Explain the different types of depreciation. 10
5. a) A small maintenance project consists of the following jobs whose precedence relationships are given below:
 Job: 1-2, 1-3, 2-3, 2-5, 3-4, 3-6, 4-5, 4-6, 5-6, 6-7
 Duration: 15 15 3 5 8 12 1 14 3 14
 (days)
 i) draw an arrow diagram
 ii) find the total float for each activity
 iii) find the critical path and project duration. 10
 b) Differentiate between CPM and PERT. 10
6. a) What is Gantt chart? Explain the advantages and disadvantages. 10
 b) A project consists of a series of tasks labelled A,B,...,I with the following relationships. Draw the network diagrams

A < D, E; B, D < F; C < G; B < H; F, G < I

Also find the minimum time of completion of the project, when the time in days of completion of each task is as follows.

Task	A	B	C	D	E	F	G	H	I
Time	23	8	29	16	24	18	19	4	10

- c) A CNC machine costs Rs. 30,00,000/- is estimated to serve eight years after which its salvage value is estimated to be 2,50,000/-.
 Find: i) Annual Depreciation ii) Book Value at the end of 4th & 6th year. Use Straight Line method of Depreciation. 6
 4

Unit – III

7. a) What is time value of money? 10
 b) A college student wants to buy a laptop for her engineering class and hopes to resell it after the completion of the course. She can purchase the unit at Rs.35,000 with an estimated salvage value of Rs.5,000 at the end of 4 years. For an interest rate of 14%, find the equivalent annual cost of this investment using capital recovery with return.
 Also explain why interest is charged on amounts. 10
8. a) Discuss effective rate of interest with suitable numerical exams. 10
 b) An assembly is to be made from two parts X and Y. Both parts must be turned on lathe, Y must be polished whereas 'X' need not be polished. The sequence of activity together with their predecessors is given below.

Activity	Description	Predecessor Activity
A	Open work order	----
B	Get material for X	A
C	Get material for Y	A
D	Turn X on the lathe	B
E	Turn Y on the lathe	B,C
F	Polish Y	E
G	Assemble X & Y	D,F
H	Pack	G

Draw a network diagram of activity for the project. 10